



Pontificia Universidad Católica de Chile
IA Lab



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Applications for Audio & Video

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SoundNet: Learning Sound Representations from Unlabeled Video

Yusuf Aytar ¹, Carl Vondrick ¹, Antonio Torralba ¹
NIPS, 2016

End-To-End Audiovisual Speech Recognition

Stavros Patridis ², Themis Stafylakis ³, Pingchuan Ma ², Feipeng Cai ²,
Georgios Tzimiropoulos ³, Maja Pantic ²
ICASSP, 2018

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- Task: Natural sound (acoustic scene) classification
- Motivation: The fields of object recognition, speech recognition, machine translation have been revolutionized by **massive labeled datasets** and **deep learning models**
- However, there has not yet been the same progress in natural sound understanding tasks, because of the **lack of large-scale labeled datasets**
- How can we overcome this limitation ?

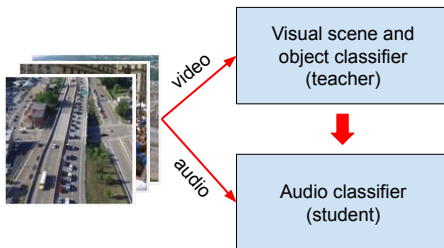
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- **Training strategy:** Student-Teacher

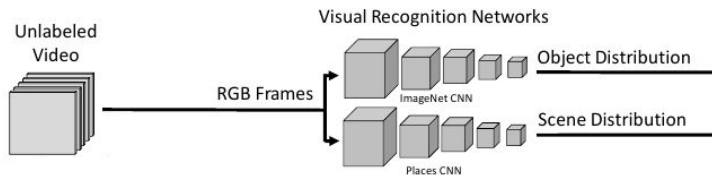


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More data allow to build deeper networks without overfitting!

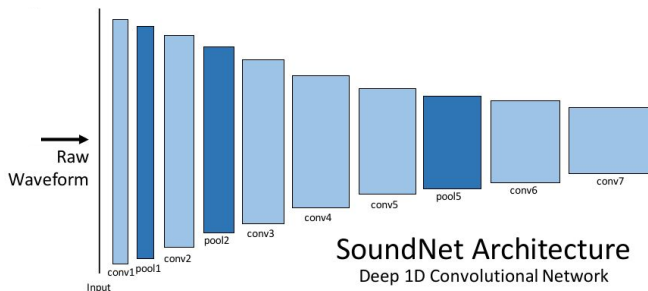
SoundNet: Teacher Model

- Teacher model makes predictions over video frames
- Two tasks have been defined for teacher model:
 - Object recognition
 - Scene recognition
- For each of the above tasks, an independent model is trained over ImageNet and Places datasets, respectively



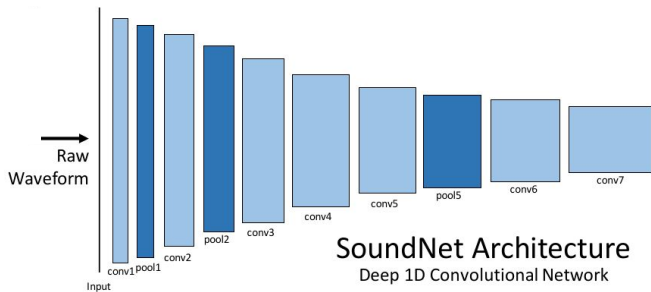
SoundNet: Student Model

- Since sound can vary in temporal length, a fully convolutional network is used



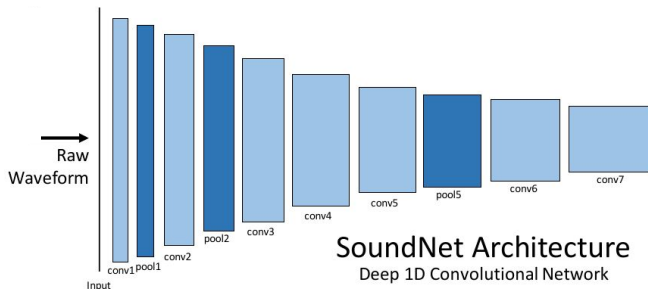
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- Since sound can vary in temporal length, a fully convolutional network is used
- Due to the representation adapts to the input length, the output layer must handle variable length inputs



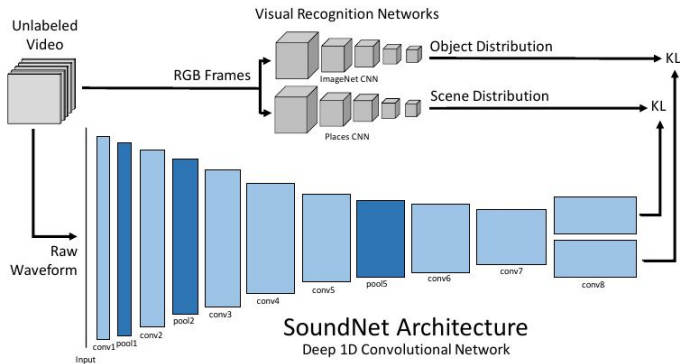
SoundNet: Student Model

- Since sound can vary in temporal length, a fully convolutional network is used
- Due to the representation adapts to the input length, the output layer must handle variable length inputs
- Since this model is trained with video, a convolutional output layer is used to produce an output over a multiple timesteps in video



SoundNet: Learning Sound Representations from Unlabeled Video

SoundNet: Student-Teacher training



SoundNet Learning

- Let $x_i \in R^D$ be a sound waveform and $y_i \in R^{3xTxWxH}$ be its corresponding video for $1 \leq i \leq N$
- During learning, they aim to use the posterior probabilities from teacher vision network $g_k(y_i)$ in order to train the student network $f_k(x_i)$ (k : number of concepts to be transferred)
- Due to the output of the vision network can be interpreted as a distribution, they use the Kullback - Leibler (KL) divergence as the loss function

$$\min_{\theta} \sum_{k=1}^K \sum_{i=1}^N D_{KL}(g_k(y_i) || f_k(x_i))$$

$$D_{KL}(P || Q) = \sum_j P_j \cdot \log\left(\frac{P_j}{Q_j}\right)$$

SoundNet Parameters

Layer	conv1	pool1	conv2	pool2	conv3	conv4	conv5	pool5	conv6	conv7	conv8
Dim.	220,050	27,506	13,782	1,722	862	432	217	54	28	15	4
# of Filters	16	16	32	32	64	128	256	256	512	1024	1401
Filter Size	64	8	32	8	16	8	4	4	4	4	8
Stride	2	1	2	1	2	2	2	1	2	2	2
Padding	32	0	16	0	8	4	2	0	2	2	0

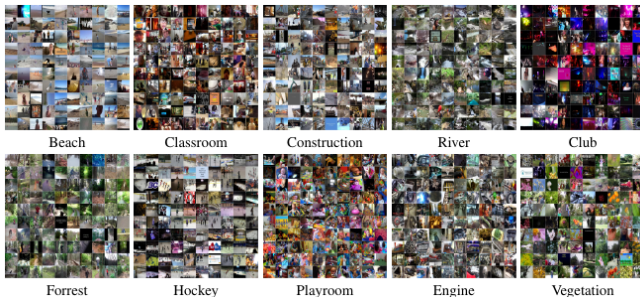
Table 1: **SoundNet (8 Layer)**: The configuration of the layers for the 8-layer SoundNet.

conv1	pool1	conv2	pool2	conv3	pool3	conv4	conv5
220,050	27,506	13,782	1,722	862	432	217	54
32	32	64	64	128	128	256	1401
64	8	32	8	16	8	8	16
2	8	2	8	2	8	2	12
32	0	16	0	8	0	4	4

Table 2: **SoundNet (5 Layer)**: The configuration for the 5-layer SoundNet.

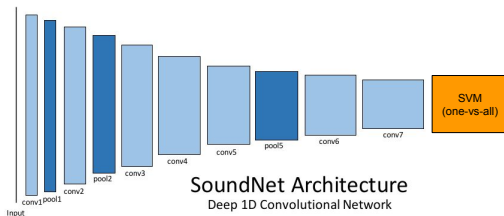
Training set

- More than 2 million of videos from Flickr by quering for popular tags
- This resulted in over one year of continuous natural sound and video for training
- The length of each video varies from a few seconds to several minutes
- Audio post-processing: convert sound to MP3, reduce sampling rate to 22 [KHz] and convert to single channel



Evaluation: Natural Sound Classification

- Although we train SoundNet to classify visual categories, the categories we wish to recognize may not appear in visual models (e.g., sneezing)
- We want to take advantage of semantic learnt from visual mode, to solve a sound classification task
- So, we will ignore the output layer and use the internal representation as features for training classifiers



Datasets for Sound Classification

- Detection and Classification of Acoustic Scenes and Events Challenge (DCASE)
 - Task: scenes from natural sounds recognition
 - 10 acoustic scene categories
 - 10 training examples per category
 - 100 held-out testing examples
- ESC-50 and ESC-10
 - Task: environmental sound recognition
 - 50 / 10 categories
 - 40 samples per category
 - sampling strategy: 5 folds with leave-one-out

Results on DCASE dataset

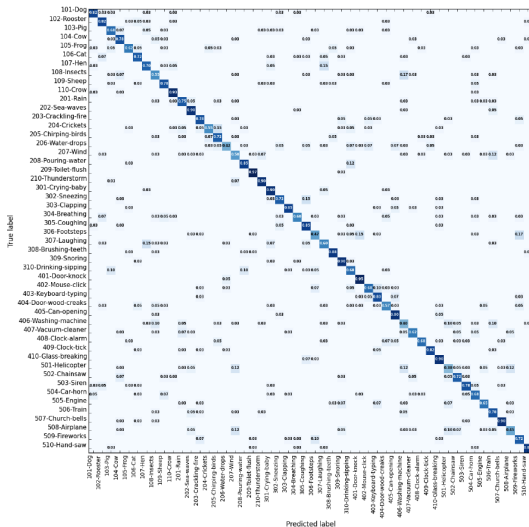
Method	Accuracy
RG [29]	69%
LTT [21]	72%
RNH [30]	77%
Ensemble [34]	78%
SoundNet	88%

Results on ESC-50 and ESC-10 datasets

Method	Accuracy on	
	ESC-50	ESC-10
SVM-MFCC [28]	39.6%	67.5%
Convolutional Autoencoder	39.9%	74.3%
Random Forest [28]	44.3%	72.7%
Piczak ConvNet [27]	64.5%	81.0%
SoundNet	74.2%	92.2%
Human Performance [28]	81.3%	95.7%

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Results on ESC-50 dataset



Ablation Analysis

Comparison of	SoundNet Model	Accuracy on	
		ESC-50	ESC-10
Loss	8 Layer, ℓ_2 Loss	47.8%	81.5%
	8 Layer, KL Loss	72.9%	92.2%
Teacher Net	8 Layer, ImageNet Only	69.5%	89.8%
	8 Layer, Places Only	71.1%	89.5%
	8 Layer, Both	72.9%	92.2%
Depth and Visual Transfer	5 Layer, Scratch Init	65.0%	82.3%
	8 Layer, Scratch Init	51.1%	75.5%
	5 Layer, Unlabeled Video	66.1%	86.8%
	8 Layer, Unlabeled Video	72.9%	92.2%

For more information visit ...

- <https://arxiv.org/abs/1610.09001>
- <http://projects.csail.mit.edu/soundnet/>

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- Goal: speech recognition from audio and video sources



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- But, why do we need video if we have the audio ?

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We need to design a model which ...

- receives audio and video streams
- returns the most probable word from a vocabulary



Introduction

Let's design our own model ...

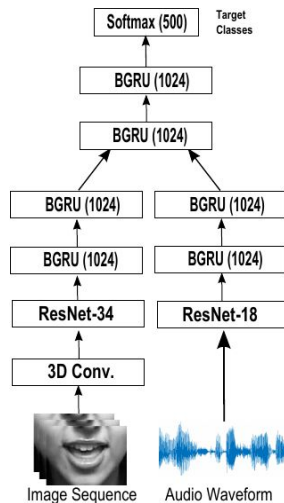
Model: main features

- This work presents an end-to-end model based on residual networks and BiGRUs
- First audiovisual fusion model using raw video and audio
- Two streams which extract features directly from mouth regions and raw waveforms
- Temporal dynamics are modeled by a 2-layer BiGRUs
- Fusion takes place via another 2-layer BiGRU

Model

- Visual stream

- Input: 29 frames
- Spatiotemporal convolution: 64, 5x7x7
- ResNet-34: drops spatial dimension
- 2-layers BiGRU



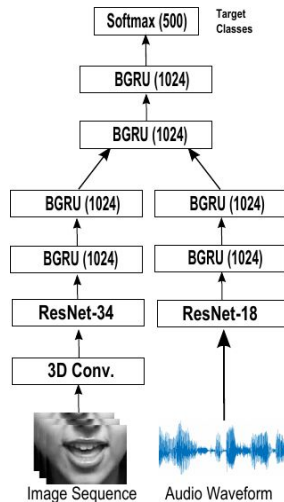
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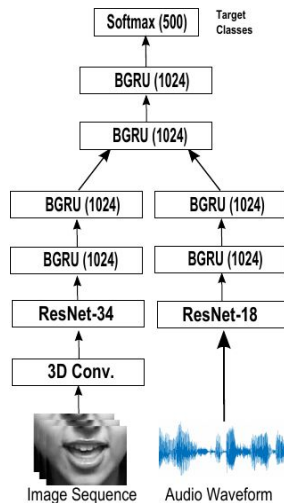
- Audio stream

- No need of spatiotemporal convolution
- ResNet-18: 1D kernels of 5 [ms], stride 0.25 [ms]
- ResNet output: 29 frames/windows using average pooling



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- Classification Layers
 - 2-layers BiGRU
 - Softmax layer provides a label to each frame



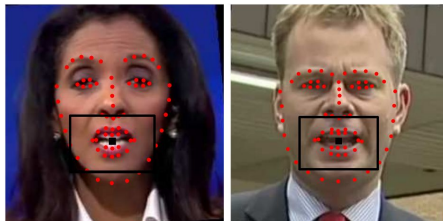
Dataset

- Lip Reading in the Wild (LRW) database ¹
- The dataset consists of short video segments from BBC programs, mainly news and talk shows
- All segments are 29 frames in length (1.16 [s])
- It contains more than 1000 speakers and large variation in head pose and illumination
- Number of words (classes): 500
- Several words are visually familiar (i.e. America and American)
- Largest publicly available lipreading dataset in the wild

¹https://www.robots.ox.ac.uk/~vgg/data/lip_reading/lrw1.html

Data Pre-processing

- Video
 - Mouth ROIs are already centered
 - A fixed bounding box of 96x96 is extracted

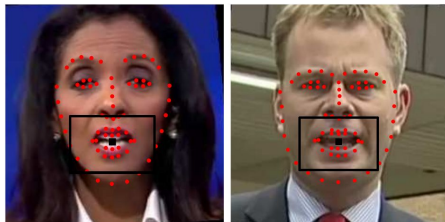


- Audio
 - z-normalized: zero mean and standard deviation one

Data Augmentation

- Video

- Random cropping
- Horizontal flip with 50% of probability (all frames)

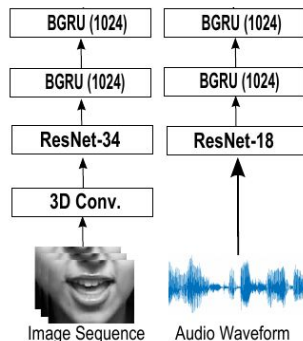


- Audio

- Random noise added at different levels: [-5 dB, 20 dB]
- NOISEX database

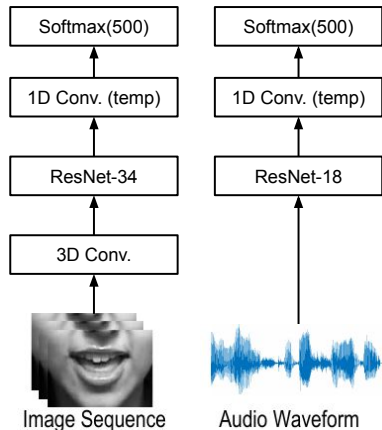
Training Process: single stream

- Each stream is trained independently
- Directly training end-to-end each stream leads to suboptimal performance



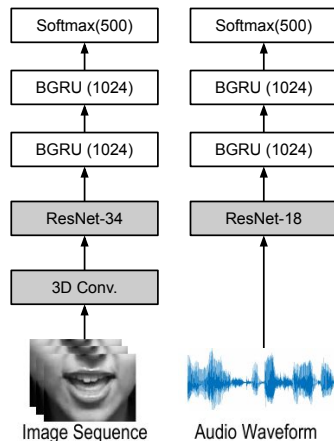
Training Process: single stream

- Each stream is trained independently
- Directly training end-to-end each stream leads to suboptimal performance
- Initially, a temporal convolutional back-end is used
- Trained until there is no improvement for more than 5 epochs in validation set



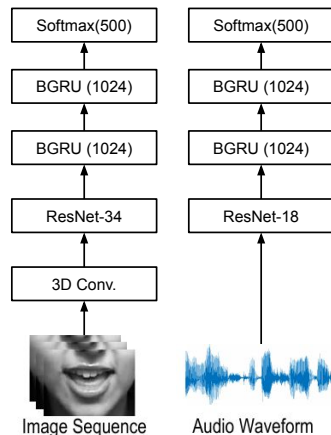
Training Process: single stream

- Temporal convolutional back-end is replaced by BiGRU
- Weights of the 3D convolutional front-end and ResNet are fixed
- BiGRU is trained for 5 epochs



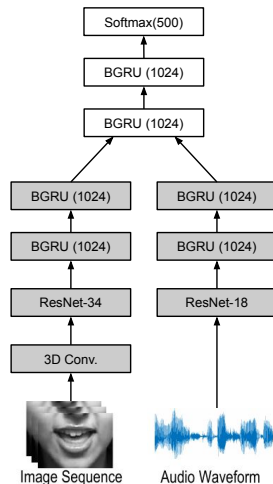
Training Process: single stream

- Once ResNet and BiGRU have been pre-trained, each stream is trained end-to-end
- Early stopping with a delay of 5 epochs is used



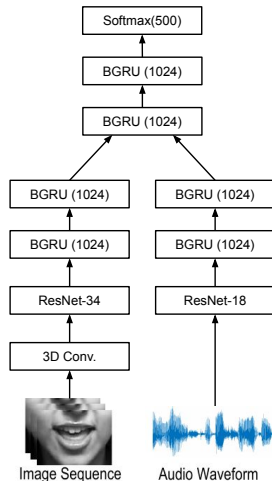
Training Process: audiovisual mode

- Once single streams have been trained, its weights are used to initialize the multi-stream architecture
- An additional BiGRU is added on top of all streams to fuse outputs
- Weights of audio and visual streams are fixed
- Top BiGRU is trained for 5 epochs



Training Process: audiovisual mode

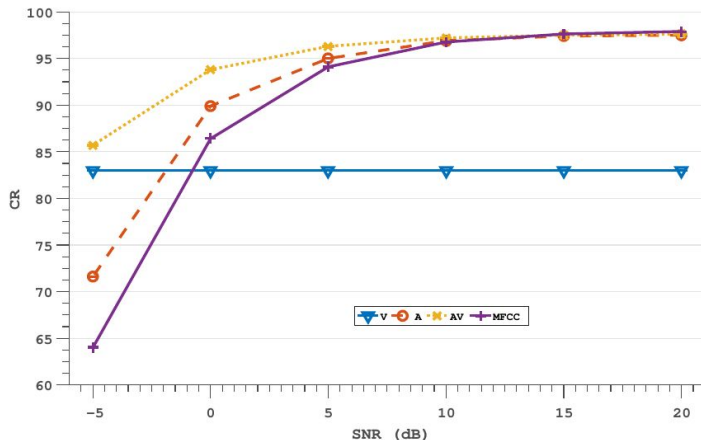
- Once top BiGRU has been pre-trained, the entire network is trained end-to-end
- Early stopping with a delay of 5 epochs is used



Results: Classification Rate

Stream	CR
A (End-to-End)	97.7
A (MFCC)	97.7
V (End-to-End)	82.0
V [13]*	83.0
V [15]	76.2
V [19]	61.1
A + V (End-to-End)	98.0

Results: Signal Noise Ratio (SNR)



Disadvantages

- It is limited to isolated and fix set of words
- Training process is very complex
- It do not generalize well to variations on sequence length



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